

Classification II

Discriminative vs Generative Models

Discriminative Models

- **directly** look at **estimating the likelihood** of y given x

$$P(Y|X)$$
- has **no knowledge** of the **distribution** of X
- **cannot generate synthetic** instances $P(X, Y)$
- these are the **models** we have studied in **earlier chapters**
 - **linear** regression, **logistic** regression

Direct

Generative Models

- models the **joint distribution** between x and y

$$P(X, Y)$$

$$P(Y|X) = \frac{P(X, Y)}{\int P(X, Y) dY}$$
- **can generate** instances of (X, Y)

Indirect

Optimal Classifier

- **Bayes Rule**

$$P(X|Y) = \frac{P(X|Y)P(Y)}{P(X)}$$

$$P(Y = y|X = x) = \frac{P(X = x|Y = y)P(Y = y)}{P(X = x)}$$

- **optimal classifier**

$$f^*(x) = \underset{Y=y}{\operatorname{argmax}} P(Y = y|X = x)$$

$$f^*(x) = \underset{Y=y}{\operatorname{argmax}} P(X = x|Y = y)P(Y = y)$$

Discriminative approach

applying Bayes Rule (Generative)

- class **prior**: tells you the **likelihoods without any knowledge of input**

Class Conditional Density

Class Prior

- class **conditional**: tells **distribution of input features given the label y**

Generative Approach with Gaussians

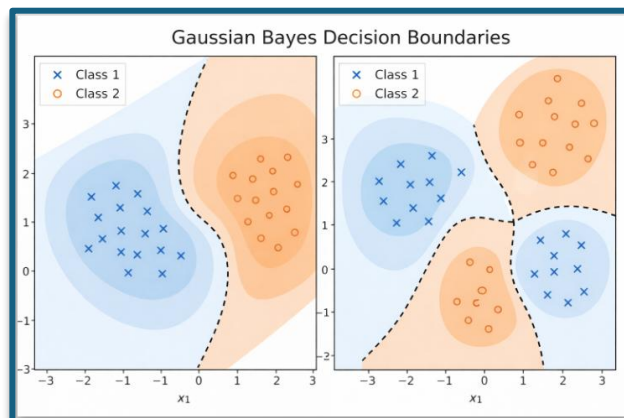
Gaussians

- we can now **consider** appropriate **models** for the two terms
 - class **probability** $P(Y = y)$
 - **Bernoulli**(θ)
 - $P(Y = \text{heads}) = \theta \quad P(Y = \text{tails}) = 1 - \theta$
 - class **conditional** distribution of $P(X = x | Y = y)$
 - **Gaussian** class **conditional** (two dimensions)
 - $P(X = x | Y = y) = \frac{1}{\sqrt{2\pi}|\Sigma_y|} \exp\left(-\frac{(x-\mu_y)\Sigma_y^{-1}(x-\mu_y)^T}{2}\right)$
 - parameters are **dependant** on the **values** we are **conditioning**
 - for **Gaussian**, **mean** and **covariance** are **dependant** on **y** value

Gaussian Bayes Classifier

- classifier that **models** the **joint likelihood** using class condition distributors by **esitimating** the **marginal distribution** of **y**
 - gives **probability** for of **y** for **all** potential **values**
- then, we **model** **x** given **y** by **fitting a Gaussian**
 - we are **varying** the **mean** and **covariance** matrix **depending** on **y**
 - each **y** gets its own **mean** and **covariance matrix**

Decision Boundary of Gaussian Bayes



the **decision boundary** can take **different forms** (generally **quadratic** for **Gaussians**)

Gaussian Bayes Parameters	- as stated, for each potential value of y - estimating probability - estimating Gaussian - estimating conditional Gaussian parameters $\{(x_i, y_i)\}_{i=1}^n$ $y = 1$ $y = 2$... $y = c$ μ_{y1}, ϵ_{y1} μ_{y2}, ϵ_{y2} μ_{y3}, ϵ_{y3} data is split into c “buckets” given c potential values estimate mean and covariance parameters for each		
How Many Parameters are Needed to Learn?	class probability $P(Y = y) = p$ for all y in $0, 1, 2, \dots, k$ $p_0, p_1, \dots, p_k$ all sum to 1 $k - 1$ parameters needed if k labels class conditional distribution of features $P(X = x Y = y) \sim N(\mu_y, \epsilon_y) \text{ for each } y$ $\mu_y - d \cdot \text{dim vector}$ $\epsilon_y - d \times d \text{ matrix}$		
Discrete Features			
Binary Classification	example: black and white image - each image is represented as a vector of d binary features - black=1, white=0 training data - input, X <table border="1"> <tr> <td>1</td> <td>2</td> </tr> </table> n images - label, Y 1 2 n labels	1	2
1	2		
Discrete Bayes Model	$P(Y = y) = p_y,$ for all $y \in \{0, 1, 2, \dots, 9\},$ $\sum_{\{y=0\}^9_y} p = 1$ $P(X = x Y = y)$ ~ For each label y, maintain a probability table with		

	$2^d - 1$ entries
Getting Rid of Parameters	
What is wrong with too many parameters?	- even for a single parameter, to build an accurate representation, you need hundreds, or even thousands, of training samples generative models often use 10^x parameters where $x > 10000$
Conditional Independence	X is conditionally independent of Y given Z: - Probability distribution governing X is independent of the value of Y, given the value of Z $(\forall x, y, z) P(X = x Y = y, Z = z) = P(X = x Z = z)$ equivalent to: $P(X, Y Z) = P(X Z) P(Y Z)$ - for example: $P(\text{Thunder} \text{Rain}, \text{Lightning}) = P(\text{Thunder} \text{Lightning})$
Naïve Bayes Classifier	Bayes Classifier with additional “naive” assumption $P(X_1, X_2 Y) = P(X_1 X_2, Y) P(X_2 Y) = P(X_1 Y) P(X_2 Y)$ $P(X_1, \dots, X_d Y) = \prod_{i=1}^d P(X_i Y)$ - if conditional independence assumption holds, Naïve Bayes (NB) is an optimal classifier (but worse otherwise). Bayes classifier with additional “naive” assumption $f_{NB}(x) = \operatorname{argmax}_y P(x_1, \dots, x_d y) P(y)$ $= \operatorname{argmax}_y \prod_{i=1}^d P(x_i y) P(y)$

Note: This does *not* mean thunder is independent of rain!

$$X = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_d \end{bmatrix}$$

features are independent given class:

has **fewer parameters**, so **less training data** even though **assumption** may be **violated**

Parameters Needed for Bernoulli Conditionals	- how many parameters? - class probabilities: $Y \in \{1, 2, \dots, K\}$ $P(Y = y) = p_y \text{ for all } y$ $K - 1$ free parameters if K labels - class conditional distribution of features (Naive Bayes assumption): $P(X_i = x_i Y = y)$ linear instead of exponential in d!
Naïve Bayes Algorithm: Discrete Features	training data: $\{(x^j, y^j)\}_{j=1}^n \quad X^j = (X_1^j, \dots, X_d^j)$ maximum likelihood estimates - class probability: $P(y) = \frac{\#\{j : Y^j = y\}}{n}$ - class conditional distribution: $\frac{P(x_i, y)}{P(y)} = \frac{\#\{j : X_i^j = x_i, Y^j = y\}}{\#\{j : Y^j = y\}}$ Naive Bayes Prediction (Test Data) $X = (x_1, \dots, x_d)$ $Y = \operatorname{argmax}_y P(y) \prod_{i=1}^d \frac{P(x_i, y)}{P(y)}$ $2Kd$ free parameters if K labels
Gaussian Conditionals	how many parameters $P(X_i = x_i Y = y) \sim N(\mu_{iy}, \sigma_{iy}^2)$ for each class y and each feature (pixel) i.

Class Probability (putting it all together)	
Class Conditional Distribution	$P(Y = y) = p_y$ <i>Free parameters: $K - 1$</i> Gaussian Naïve Bayes $P(X_i = x_i Y = y) \sim N(\mu_{iy}, \sigma_{iy}^2)$ - for each class y and each feature i: <i>Parameters per class – feature pair: 2</i> <i>Total class – conditional parameters: $2Kd$</i> Total Number of Parameters $(K - 1) + 2 K d$
Generative vs Naive Generative	Generative Model for Classification $P(X_i = x_i Y = y) \quad P(Y = y)$ Naïve Bayes Algorithm $\prod_{i=1}^d P(X_i = x_i Y = y) \quad P(Y = y)$
Case Study	Classify e-mails $Y = \{\text{Spam, NotSpam}\}$ Classify news articles $Y = \{\text{topic of the article}\}$ Classify webpages $Y = \{\text{Student, Professor, Project, ...}\}$ - What about the features X? <i>The text!</i>

Summary

- **Optimal decision** using **Bayes classifier**
- **Naïve Bayes** classifier
 - What's the **assumption**
 - **Why** we **use** it
 - **How** do we **learn** it
- **Gaussian NB**
 - **Features** are still **conditionally independent**
 - Each **feature** has a **Gaussian distribution** given class
- **Text classification**
 - Bag-of-words model